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1 /Users/mehakagrawal/PycharmProjects/
  Final_Dissertation/.venv/bin/python /Users/
  mehakagrawal/Desktop/Final_Dissertation/Codes/10.
  grn_interaction_pattern_mapping/
  grn_interaction_pattern_mapping.py
2 Mapping the GRN onto the interaction model patterns
3
4 Interaction genes with a human orthologue: 14488 (
  from 21008)
5 /Users/mehakagrawal/Desktop/Final_Dissertation/
  Codes/10. grn_interaction_pattern_mapping/
  grn_interaction_pattern_mapping.py:119:
  FutureWarning: Downcasting object dtype arrays on .
  fillna, .ffill, .bfill is deprecated and will
  change in a future version. Call result.
  infer_objects(copy=False) instead. To opt-in to the
  future behavior, set `pd.set_option('future.
  no_silent_downcasting', True)`
6 per_gene['is_grn_target'] = is_target.reindex(
  per_gene.index).fillna(False)
7 GRN target genes: 1399 (of 14488 orthologue-mapped
  genes)
8 Age-dependent onset: 68.3% of GRN targets vs 34.6%
  background
9 Fisher exact (targets enriched for onset): odds
  ratio = 4.81, p = 4.52e-162
10 GRN targets with significant interaction: 44.8% vs
  22.5% background
11
12 Interaction pattern of GRN targets:
13 interaction_pattern
14 age_dependent_onset      956
15 no_genotype_effect      441
16 transient_early          2
17
18 Direction at 18mo of GRN targets:
19 direction_18mo
20 down_in_MUT      806
21 up_in_MUT        593
22
23 Saved: grn_target_interaction_map.csv,

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23 grn_tf_pattern_summary.csv, grn_pattern_enrichment.  
   png  
24  
25 Outputs in: /Users/mehakagrawal/Desktop/  
   Final_Dissertation/Outputs/10.  
   grn_interaction_pattern_mapping  
26  
27 Process finished with exit code 0  
28
```